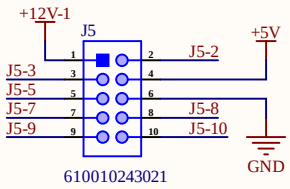
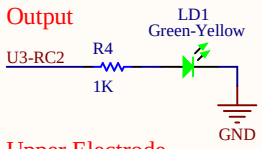
SSR-05D-PIC-2-1.6 (PCB 2)

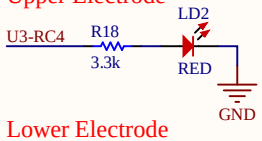
Female Header



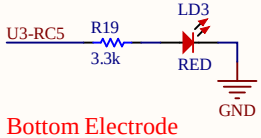
Output



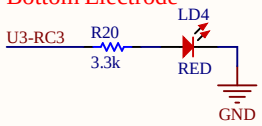
Upper Electrode



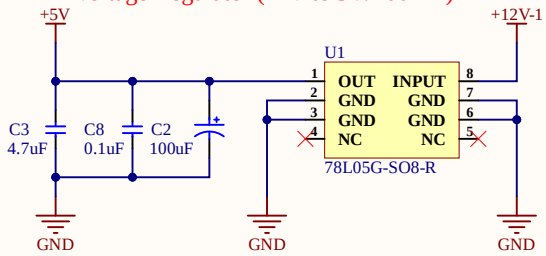
Lower Electrode



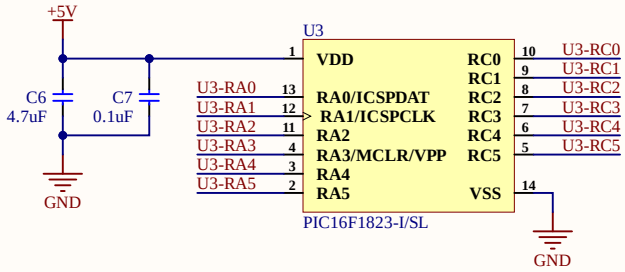
Bottom Electrode



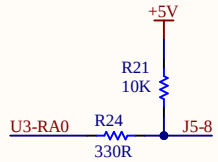
Voltage Regulator (12V to 5V/100mA)



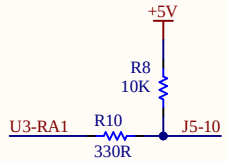
PIC Microcontroller



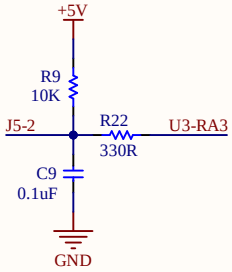
Sensing Circuit (Lower Electrode)



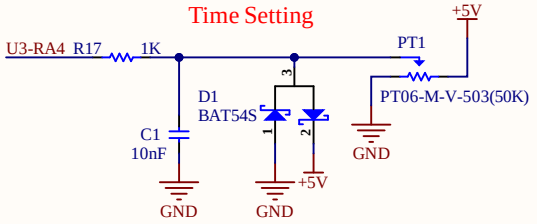
Sensing Circuit (Upper Electrode)



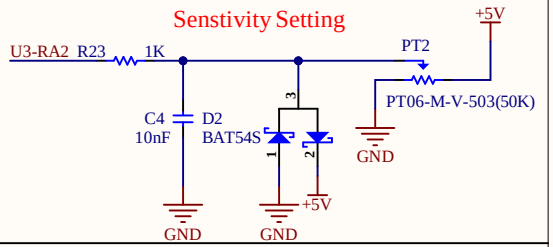
Additional Circuit



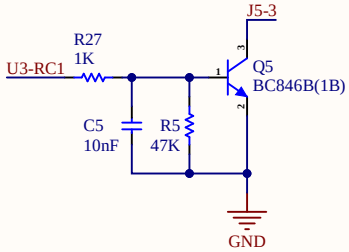
Time Setting



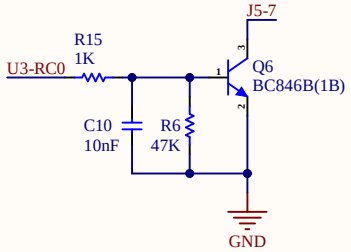
Senstivity Setting



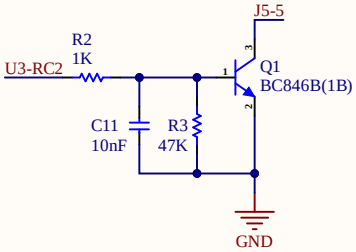
Upper Electrode Control Circuit



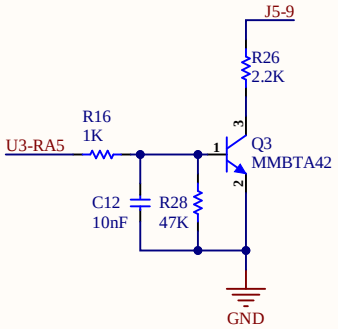
Lower Electrode Control Circuit



Relay Control Circuit



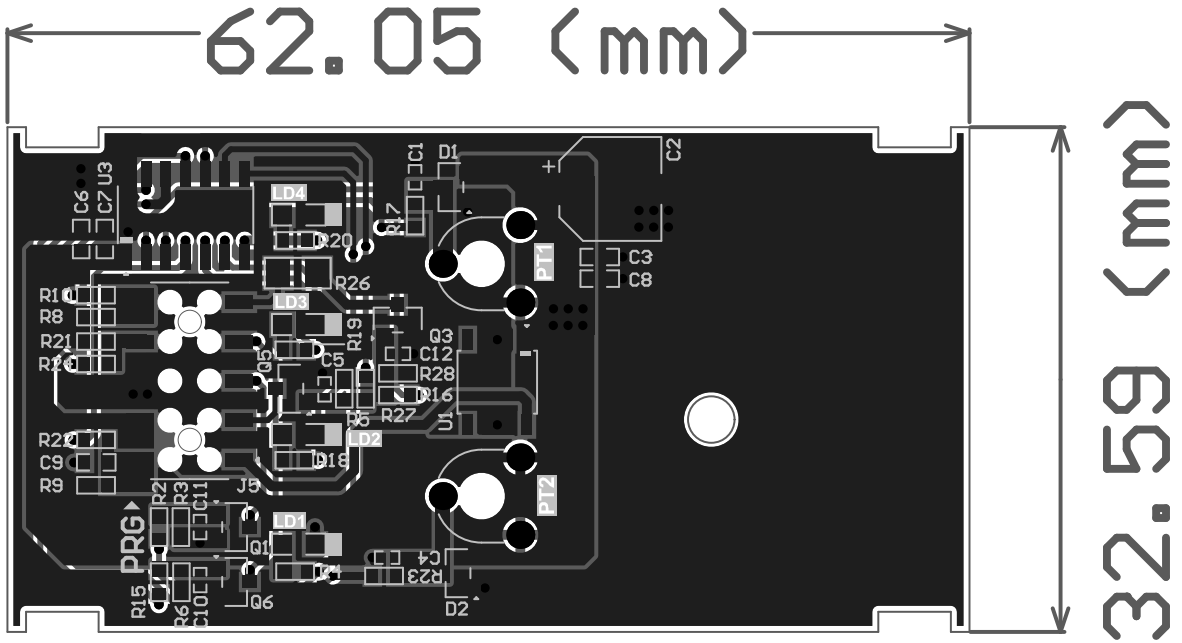
Bottom Electrode Control Circuit



eSpark Consulting Group

Title: LLR-05U (Secondary-Board-Schematic).S	
File: LLR-05U (Secondary-Board).PrjPcb	
Date: 04/07/2024	Page Num: 1
Drawn By: Huzaifa Ali	Page Total: 1

Layer	Name	Material	Thickness	Constant	Board Layer Stack
	Top Overlay				
	Top Solder	Solder Resist	0.076mm	3.5	
1	Top Layer		0.254mm		
	Dielectric 1	FR-4	0.940mm	4.8	
2	Bottom Layer		0.254mm		
	Bottom Solder	Solder Resist	0.076mm	3.5	
	Bottom Overlay				
Total board thickness:			1.600mm		



Comment	Description	Designator	Footprint	LibRef	Quantity
10nF	CAP CER 10000PF 25V X7R 0402	C1, C4, C5, C10, C11, C12	Cap-0402	CC0402KRX7R8BB103	6
100uF	100uF 10V 55mA@120Hz ±20% SMD,D6.3xL5.4mm Aluminum Electrolytic Capacitors - SMD ROHS	C2	RVT1A101M0605	RVT1A101M0605	1
4.7uF	CAP CER 4.7UF 25V X5R 0603	C3, C6	Cap-0603	GRM188R61E475KE11D	2
0.1uF	CAP CER 0.1UF 50V X7R 0603	C7, C8, C9	Cap-0603	CL10B104KB8WPNC	3
BAT54S	30V Dual 200mA 800mV@100mA SOT-23 Schottky Barrier Diodes (SBD) ROHS	D1, D2	SOT-23	BAT54S	2
610010243021	SMT Vertical Socket Header WR-PHD, Pitch 2.54 mm, Dual Row, 10 pins	J5	610010243021	610010243021	1
Green-Yellow	0805 普绿色 (Iv=26-75mcd(Typ) @IF=5mA)	LD1	LED-0805	XL-2012SYGC	1
RED	20mA 无色透明透镜 67mcd~195mcd -30℃~+85℃ 615nm~630nm 独立二极管 Red 130° 65mW 1.6V~2.6V 0805 Light Emitting Diodes (LED) ROHS	LD2, LD3, LD4	LED-0805	NCD0805R1	3
PT06-M-V-503(50K)	20% 150mW 50kΩ - Variable Resistors/Potentiometers ROHS	PT1, PT2	PT06-M-V-503(50K)	PT06-M-V-503(50K)	2
BC846B(1B)	65V 300mW 200@2mA,5V 100mA NPN SOT-23 Bipolar Transistors - BJT ROHS	Q1, Q5, Q6	SOT-23	BC846B(1B)	3
MMBTA42	300V 350mW 100@10mA,10V 300mA NPN SOT-23 Bipolar Transistors - BJT ROHS	Q3	SOT-23	MMBTA42	1
1K	RES SMD 1K OHM 0.1% 1/10W 0603	R2, R4, R15, R16, R17, R23, R27	Res-0603	RT0603BRD071KL	7
47K	RES SMD 47K OHM 0.1% 1/10W 0603	R3, R5, R6, R28	Res-0603	RT0603BRD0747KL	4
10K	RES SMD 10K Ohm 1% 1/10W 0603	R8, R9, R21	Res-0603	RC0603FR-0710KL	3
330R	100mW Thick Film Resistors 75V ±100ppm/℃ ±1% 330Ω 0603 Chip Resistor - Surface Mount ROHS	R10, R22, R24	Res-0603	0603WAF3300T5E	3
3.3k	RES SMD 3.3K OHM 0.1% 1/10W 0603	R18, R19, R20	Res-0603	RT0603BRD073K3L	3
2.2K	250mW Thick Film Resistors 200V ±100ppm/℃ ±1% 2.2kΩ 1206 Chip Resistor - Surface Mount ROHS	R26	Res-1206	1206W4F2201T5E	1
78L05G-SO8-R	Pos 5V 0.1A 8-Pin SOP T/R	U1	SOP-8	78L05G-SO8-R	1
PIC16F1823-I/SL	IC MCU 8BIT 3.5KB FLASH 14SOIC	U3	SOIC-14	PIC16F1823-I/SL	1